

Introduction to Video Rec

Defination- Systems designed to suggest videos to users based on various factors, enhancing user engagement and satisfaction

Summary

- **MicroLens Dataset**

- **Scale:** 1 billion interactions from 34 million users and 1 million micro-videos
- **Data:** Includes video titles, cover images, audio, and full-length videos
- **Purpose:** Multimodal recommendation research and short-video personalization

- **MovieLens Dataset**

- **Popular Use Case:** Movie recommendations
- **Data:** Millions of ratings and tags for movies
- **Versions:** Available in multiple sizes (100K, 1M, 10M ratings)
- **Strength:** Widely used benchmark for collaborative filtering algorithms

- **. KuaiRand Dataset**

Outline

- Business Problem
- Data
- Methods
- Results
- Conclusions

Business Problem

- **User Engagement and Retention:** By providing personalized recommendations, platforms keep users engaged longer and reduce churn.
- **Revenue Growth:** Accurate targeting increases ad revenue and conversion rates.
- **User Satisfaction:** Recommenders enhance the user experience by delivering relevant and diverse content.
- **Content Discovery:** They expose users to long-tail content, maximizing content utilization.
- **Competitive Advantage:** Personalized experiences help platforms stand out in a competitive market.
- **Cost Efficiency:** Insights guide content production and acquisition, reducing investment risks.

Data

- **Large-Scale Interactions:** Datasets contain millions to billions of user-video interactions, capturing real-world user behavior.
- **Multimodal Data:** Include diverse data types like text (titles), images (thumbnails), audio, and full-length videos (e.g., **MicroLens**).
- **Rich Metadata:** Provide detailed user and video metadata, enabling personalized recommendations (e.g., **KuaiRec**).
- **Implicit and Explicit Feedback:** Record both implicit interactions (watch time, clicks) and explicit ratings/tags (e.g., **MovieLens**).
- **Sequential and Random Exposure:** Include sequential user interactions and unbiased random exposures for better model evaluation (e.g., **KuaiRand**).
- **Real-World Context:** Collected from popular video platforms like Kuaishou and MTS Kion, reflecting real user preferences and trends.

Methods

- **Data Preparation:** Clean and preprocess data, handle missing values, and normalize features. Split data into training, validation, and test sets.
- **Analysis:** Perform exploratory data analysis (EDA) to understand user behavior and interaction patterns. Use visualization tools to identify trends.
- **Modeling:** Apply collaborative filtering, content-based filtering, and deep learning models

Results

- **Improved Recommendations:** More accurate and personalized video suggestions, increasing user engagement.
- **Higher User Retention:** Enhanced user satisfaction and reduced churn rates.
- **Revenue Growth:** Better ad targeting and increased conversion rates.
- **Diverse Content Discovery:** Balanced exposure to popular and long-tail videos.

Conclusions

Video recommender system datasets enable the development of accurate, personalized recommendation models. By leveraging rich user interaction data and multimodal features, the models enhance user engagement, satisfaction, and revenue growth. They also improve content discovery and provide a competitive advantage. These datasets are crucial for building effective, user-centric video platforms.

Thank You!

Email: jacklinewaceke199@gmail.com

GitHub: @JacklineWaceke

LinkedIn: [linkedin.com/Jackline Waceke/](https://www.linkedin.com/Jackline Waceke/)